

Trainings ▾

General Information

with Mechanically Actuated Injector

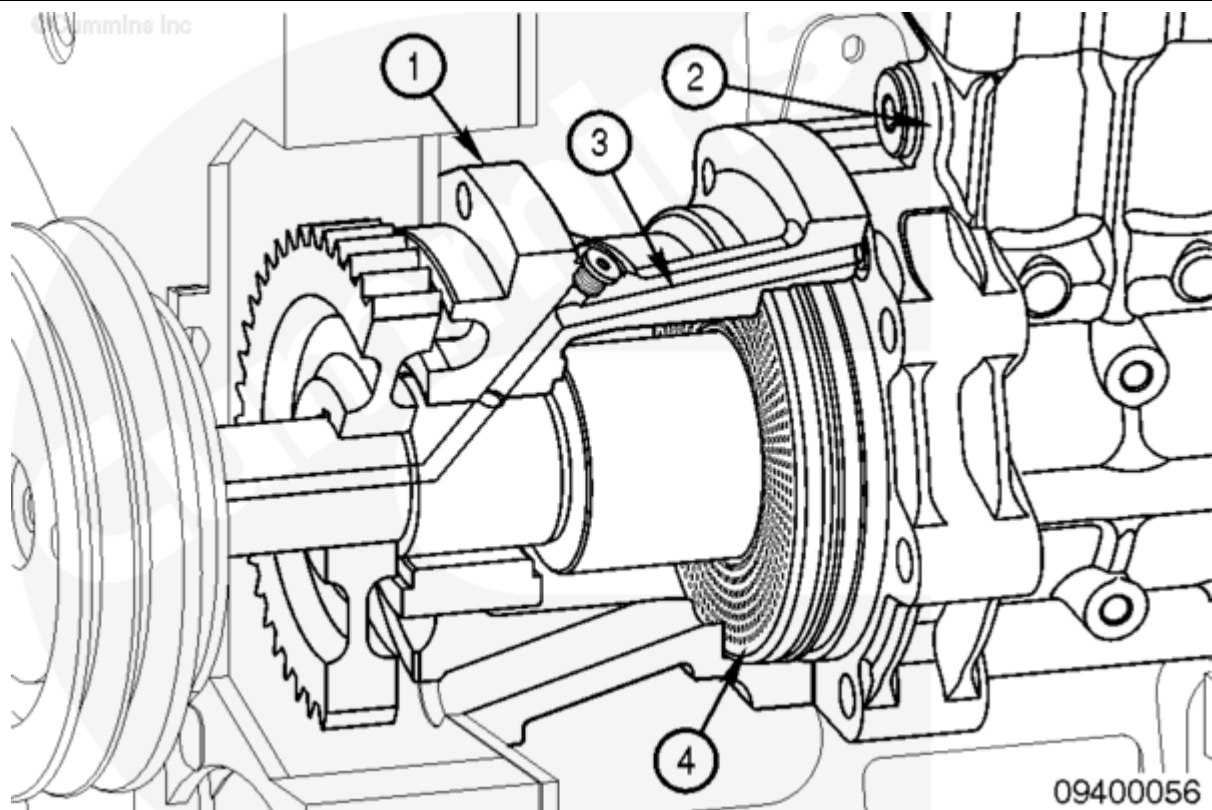
CAUTION

Installing an air compressor on a fuel pump drive housing will cause the air compressor to fail due to lack of lubrication.

The fuel pump housings are available in two designs, with and without provisions for mounting an air compressor. Those without air compressors do **not** have provisions for an air compressor oil feed line or a drain back line. The fuel pump drive has provisions for a lovejoy coupling. The compressor drive has provisions for a splined sleeve type coupling. The procedures are identical for both of the designs.

The retainer capscrew on the compressor drive (splined half-coupling) is special; it **must** have a drilling in it to provide lubrication for the splined coupling. The retainer capscrew on the fuel pump drive (lovejoy half-coupling) does **not** have an oil drilling.

with Electronically Actuated Injector



The fuel pump drive (1) located on the back side of the accessory drive housing drives the fuel pump (2). The fuel pump engages the fuel pump drive with splines cut on the end of the fuel pump shaft engaging mating splines in the back of the fuel pump drive.

The fuel pump drive also has a oil passage (3) to transfer engine oil to the fuel pump for lubrication.

A mesh (4) is installed in the fuel pump drive on the mating face to the fuel pump.

This fuel pump does **not** have any provisions for supporting an air compressor.

General Information

A hydraulic pump can be mounted on the front or rear of the gear cover. The SAE A two bolt type of flange is required to mount a hydraulic pump at the rear of the gear cover. The SAE B four-bolt type of flange is required for a hydraulic pump to be mounted on the front of the gear cover. There is a specific drive required to match the different sizes of spline drives required for optional hydraulic pumps.

Engines that do **not** have a hydraulic pump drive **must** have parts that block the oil holes in the gear cover and housing for correct engine oil pressure. A four bolt flange that has a blind plug welded to it **must** be installed on the front of the gear cover. The blind plug will extend into the bushing, blocking the oil drilling. A three bolt cover that has two o-rings **must** be installed on the rear of the gear housing.

A hydraulic pump drive can be installed by removing the blind plug flange and cover. If **only** the rear gear drive port is used, a plain four bolt cover and gasket **must** be installed on the front of the cover. If **only** the front drive port is used, a two bolt cover and gasket **must** be installed on the rear of the pump drive.

All QSK45 and QSK60 engines are equipped with an engine barring device. The purpose of this device is to provide a method of barring the engine to adjust the valves and injectors. The barring device is provided because the capscrews on the front of the crankshaft are **not** capable of withstanding the force required to rotate the engine and are **not** accessible in many of the applications. The QSK45 and QSK60 engines have marks located on the flywheel and flywheel housing that are used to position the engine for adjusting the valves and injectors. The marks are located near the barring device mounting location. Some engines have two barring devices.

The barring device contains a gear that engages the flywheel ring gear. The device also contains a spring that will push the barring device gear from the flywheel ring gear when the engine is operating. An external cotter clip is used to prevent the barring device gear and shaft from moving and contacting the ring gear when the engine is in operation.

Engines that contain a wet type flywheel housing **must** have a gasket installed between the barring device and the flywheel housing to prevent oil leaks. Engines with a dry type flywheel housing do **not** require this gasket.